



COMPETENCY STANDARD
FOR
INDUSTRIAL PRODUCTIVITY IMPROVEMENT

(LEATHER GOODS & FOOTWEAR SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Industrial Productivity Improvement** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document was originally developed under the **Skills for Employment Investment Program (SEIP)** and later reviewed and updated with the input from occupation-specific industry experts to be utilized in training initiatives under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. The document is owned by the Finance Division, Ministry of Finance, People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (30 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LET-IPI-01-G	Carry out workplace interaction	1. Interpret workplace communication and etiquette 2. Read and understand workplace documents 3. Participate in workplace meetings and discussions 4. Practice professional ethics at work	15
SICIP-LET-IPI-02-G	Operate in a team environment	1. Identify team goals and work processes 2. Identify own role and responsibilities within team 3. Communicate and co-operate with team members 4. Practice problem solving within the team	15
Total Hour			30 Hrs.

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LET-IPI-01-S	Carry out measurements and calculations	1. Plan and prepare 2. Obtain measurements 3. Perform calculations	10
SICIP-LET-IPI-02-S	Apply green practices in LET sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			20 Hrs.

Occupation Specific Competencies (250 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LET-IPI-01-O	Apply Work Processes for improving Productivity	1. Identify key production stages in leather and footwear manufacturing. 2. Assess current workflow and identify bottlenecks. 3. Record process timings and material flow. 4. Analyze causes of production delays and waste. 5. Interpret areas for improvement.	20
SICIP-LET-IPI-02-O	Apply Basic Industrial Engineering (IE) activities	1. Identify different types of line layout. 2. Perform work study process. 3. Prepare skills matrix for production line 4. Implement line balancing in production. 5. Develop value stream mapping. 6. Prepare efficiency report	70
SICIP-LET-IPI-03-O	Apply Lean Manufacturing techniques	1. Identify 3M (muda, muri and mura) 2. Identify seven types of waste in production process. 3. Apply 5S techniques in workplace. 4. Implement lean tools 5. Implement continuous improvement (Kaizen) activities.	40
SICIP-LET-IPI-04-O	Optimize Machine and Equipment Utilization	1. Identify machines in the production line. 2. Record machine downtime and its causes. 3. Schedule regular inspection. 4. Implement quick changeover (SMED) techniques.	40
SICIP-LET-IPI-05-O	Improve	1. Identify skill levels of	30

Code	Unit of Competency	Elements of Competency	Duration (hours)
	Workforce Efficiency	workers. 2. Set realistic production targets. 3. Assign tasks based on worker skills capability. 4. Monitor performance and provide feedback.	
SICIP-LET-IPI-06-O	Implement Quality Control and Assurance Techniques	1. Identify common defects. 2. Inspect materials and products. 3. Apply standard operating procedures (SOPs). 4. Record and report quality deviations.	20
SICIP-LET-IPI-07-O	Monitor and Evaluate Productivity Performance	1. Collect data on production. 2. Calculate key productivity indicators. 3. Compare performance against set targets. 4. Identify causes of underperformance. 5. Prepare reports and suggest improvement actions.	30
Total Hours			0 Hrs.

COMPETENCY STANDARD: INDUSTRIAL PRODUCTIVITY IMPROVEMENT

Generic Competencies

UNIT OF COMPETENCY: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 15 hrs.	Unit Code: SICIP-LET-IPI-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the task of interpreting workplace communication and etiquette, Reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1. Workplace documents are interpreted correctly. 2.2. Visual information/symbols/signage are understood correctly and followed. 2.3. Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4. Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1. Team meetings are attended on time. 3.2. Meeting procedures and etiquette are followed. 3.3. Active participation is ensured, opinions are expressed and heard. 3.4. Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.6 Notes 1.7 Arranging a meeting 1.8 Agenda 1.9 Simple reports such as progress and incident reports 1.10 Job sheets 1.11 Operational manuals 1.12 Brochures and promotional material 1.13 Visual and graphic materials 1.14 Standards 1.15 OHS information 1.16 Signs
3. Appropriate sources	1.17 Human Resources (HR) Department 1.18 Managers 1.19 Supervisors 1.20 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace codes of conduct 1.2 Workplace communication and etiquette 1.3 Workplace documents, signs and symbols 1.4 Meeting procedure and etiquette 1.5 Professional ethics 1.6 Responsibilities of team member
2. Underpinning skill	1.7 Demonstrating workplace communication and etiquette 1.8 Interpreting workplace instructions and symbols 1.9 Performing responsibilities of a team member 1.10 Demonstrating active participation in workplace meeting 1.11 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.1 Workplace (simulated or actual)

	1.2 Projector 1.3 Learning manual
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings 1.4 Performed responsibilities of a team member
2. Methods of assessment	Competency should be assessed by: 1.5 Written test 1.6 Practical Demonstration 1.7 Oral Questioning 1.8 Portfolio (Optional)
3. Context of assessment	1.9 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.10 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 15 hrs.	Unit Code: SICIP-LET-IPI-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the task of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team members and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members are identified and interpreted.
2. Identify own role and responsibilities within team	2.1 Personal role and responsibilities are identified within the team environment. 2.2 Reporting relationships are interpreted within team and external to team.
3. Communicate and cooperate with team members	3.1 Other teammates' tasks are identified and support provided when requested. 3.2 The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3 Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1 Problems faced at the individual and team level are identified and showed insight into the root causes of the problems. 4.2 A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3 The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4 It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
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1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Phone directory 1.8. Policy, procedure and standards 1.9. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of team members 1.3. Personal role and responsibilities 1.4. Sharing information 1.5. Views and opinions 1.6. Problem solving within the team
2. Underpinning skill	1.1. Identifying team goals and work processes 1.2. Identifying own role and responsibilities within team 1.3. Communicating and cooperating with team members 1.4. Practicing problem solving within the team
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.3 communicated and cooperated with team members 1.4 practiced problem solving within the team
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2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Sector Specific Competencies

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 10 hrs.	Unit Code: SICIP-LET-IPI-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for carry out measurements and calculations. It specifically includes the task of planning and preparing, obtaining measurements and performing calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan and prepare	1.1. Work instructions are confirmed and applied to the job in hand. 1.2. Materials to be measured are identified as per job specification. 1.3. Appropriate <u>measuring device</u> is identified and selected based on materials to be measured. 1.4. Specifications are obtained and verified from relevant <u>documents</u> .
2. Obtain measurements	2.1 Method of obtaining measurement is selected and applied. 2.2 <u>Measurements</u> are obtained using appropriate device in accordance with job requirement. 2.3 Measurements, including area, volume, tolerance and clearance limits, are confirmed and applied
3. Perform calculations	3.1 <u>Calculations</u> , using basic operations, for determining material requirement are taken. 3.2 Appropriate <u>formulas</u> for calculating quantities are selected. 3.3 Quantities are estimated from the calculation taken. 3.4 Material quantities are calculated, confirmed and recorded within tolerances.

Range of Variables

Variable	Range
	May include but not limited to:
1. Measuring device	1.1 Measuring tape 1.2 Steel rule 1.3 Calculator 1.4 Sets square
2. Documents	2.1. Technical manuals

	2.2. Specifications 2.3. Sketches 2.4. Drawings 2.5. Charts 2.6. Photographs
3. Measurements	3.1. Length 3.2. Width 3.3. Weight 3.4. Tolerance
4. Calculations	4.1 Addition 4.2 Subtraction 4.3 Multiplication 4.4 Division 4.5 Area 4.6 Volume 4.7 Circumference 4.8 Cubic meter 4.9 Volumetric weight
5. Formulas	5.1 Fractions 5.2 Percentages 5.3 Mixed numbers 5.4 Conversions 5.5 Scales

Curricular Content Guide

1. Underpinning Knowledge	1.1 Measuring devices 1.2 Measurements 1.3 Simple calculation techniques 1.4 Appropriate formulas 1.5 Estimating quantities
2. Underpinning Skills	2.1 Identifying appropriate measuring devices 2.2 Carrying out measurements for appropriate device 2.3 Performing calculations using appropriate formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Map/globe 4.3 Projector 4.4 Stationery 4.5 Learning manual.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 planed and prepared 1.2 obtained measurements 1.3 performed calculations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN LET SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-LET-IP1-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in LET sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices in LET (Ready-Made Garment)</u> manufacturing are explained. 1.2 Key terms and symbols in LET production processes are identified. 1.3 <u>Sources of environmental impacts</u> during LET manufacturing activities are identified. 1.4 LET manufacturing activities use are explained. 1.5 <u>Ways of mitigating environmental impacts</u> in LET manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in LET manufacturing are documented. 2.2 <u>Recyclable and non-recyclable items</u> in LET production processes are identified. 2.3 Procedures to reduce resource consumption in LET manufacturing are implemented. 2.4 Sustainable alternatives in LET manufacturing are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste in LET manufacturing</u> are identified. 3.2 Hazardous waste in LET production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices in LET (Ready-Made Garment)	1.1 Reducing energy consumption in LET manufacturing processes. 1.2 6R for waste management in LET production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste

	generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in LET manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in LET production with low environmental impact. 1.4 Recycling of materials used in the LET manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the LET sector.
2. Sources of environmental impacts	2.1 Air Pollution in LET Manufacturing 2.2 Water Pollution in LET Manufacturing 2.3 Soil Degradation in LET Manufacturing 2.4 Noise Pollution in LET Manufacturing 2.5 Waste Generation in LET Manufacturing 2.6 Resource Depletion in LET Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in LET production processes. 3.2 Adopting renewable energy sources for LET manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in LET Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 Non-Recyclable Items in LET Manufacturing 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in LET manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production

	and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in LET (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in LET manufacturing 1.6 Hazardous waste in LET production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in LET manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in LET manufacturing 2.6 Disposing hazardous waste in LET production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts
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	1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation Specific Competencies

Unit of Competency: APPLY WORK PROCESSES FOR IMPROVING PRODUCTIVITY	Nominal Duration: 20 Hrs.	Unit Code: SICIP-LET-IP1-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply work processes for improving productivity. It specifically includes the tasks of identifying key production stages in leather and footwear manufacturing, assessing current workflow and identify bottlenecks, recording process timings and material flow, analyzing causes of production delays and waste and interpreting areas for improvement.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify key production stages in leather and footwear manufacturing.	1.1 <u>Major production processes in leather goods</u> are identified. 1.2 <u>Major production processes in footwear manufacturing</u> are identified. 1.3 <u>Products</u> are identified. 1.4 Opportunities for time and cost reduction are recognized. 1.5 Work process data are collected and documented.
2. Assess current workflow and identify bottlenecks.	1.1 <u>Current workflow</u> in the production area is interpreted. 1.2 Work activities and <u>movement of materials</u> are monitored. 1.3 Delays and idle times are identified and noted. 1.4 Bottlenecks affecting productivity are identified. 1.5 Causes of workflow interruptions are analyzed. 1.6 Observations are documented for further evaluation. 1.7 Improvement areas are highlighted for corrective action.
3. Record process timings and material flow.	3.1 Process timings at each production stage are recorded. 3.2 Movement of materials between workstations is tracked. 3.3 Start and finish times for each task are documented. 3.4 Idle and waiting times are identified and logged. 3.5 Data on material flow and handling are collected accurately. 3.6 <u>Records</u> are maintained using approved formats or tools. 3.7 Collected data are reviewed for consistency and

	accuracy.
4. Analyze causes of production delays and waste.	4.1 <u>Collected workflow and timing data</u> are analyzed systematically. 4.2 Causes of production delays are identified and categorized. 4.3 <u>Sources of material waste</u> are detected and recorded. 4.4 Inefficient work practices are examined and noted. 4.5 Data are interpreted to determine root causes of inefficiency. 4.6 Analysis results are documented for improvement planning.
5. Interpret areas for improvement.	4.1 <u>Findings from workflow analysis</u> are reviewed and interpreted. 4.2 Areas requiring process improvement are identified. 4.3 Suggestions for reducing delays and waste are developed. 4.4 Options for simplifying or combining tasks are proposed. 4.5 <u>Recommendations</u> are aligned with productivity goals. 4.6 Improvement ideas are discussed with concerned personnel. 4.7 Proposed actions are documented for management consideration.

Range of Variables

Variable	Range (Includes but not limited to):
1. Major production processes in leather goods	1.1 Material Selection 1.2 Pattern Making 1.3 Cutting 1.4 Skiving and Splitting 1.5 Edge Coloring and Creasing 1.6 Assembling 1.7 Stitching 1.8 Trimming and Cleaning 1.9 Fitting and Attachment 1.10 Finishing 1.11 Inspection 1.12 Packaging
2. Major production processes in footwear manufacturing	2.1 Material Selection 2.2 Cutting 2.3 Skiving and Splitting 2.4 Stitching/Closing 2.5 Lasting 2.6 Sole Preparation 2.7 Sole Attaching

	2.8 Finishing 2.9 Inspection 2.10 Packaging
3. Products	3.1 Shoes (formal, casual, sports) 3.2 Sandals and slippers 3.3 Boots (work, casual, safety) 3.4 Bags (handbags, backpacks, travel bags) 3.5 Wallets and purses 3.6 Belts 3.7 Gloves 3.8 Leather jackets and apparel 3.9 Accessories (watch straps, key holders) 3.10 Leather goods for home décor (cushions, organizers)
4. Current workflow	4.1 Cutting 4.2 Stitching 4.3 Lasting 4.4 Finishing 4.5 Packaging 4.6 Material handling processes
5. Movement of materials	5.1 Raw materials 5.2 Semi-finished 5.3 Finished goods. 5.4 Manual handling 5.5 Conveyors 5.6 Trolleys 5.7 Forklifts
6. Records	6.1 Log sheets 6.2 Time-study charts 6.3 Digital tools 6.4 Software
7. Collected workflow and timing data	7.1 Workflow charts 7.2 Process timings 7.3 Material movement 7.4 Production records
8. Sources of material waste	8.1 Cutting waste 8.2 Defective products 8.3 Scrap 8.4 Excess inventory
9. Findings from workflow analysis	9.1 Data on process timings 9.2 Material flow 9.3 Bottlenecks 9.4 Idle times 9.5 Waste
10. Recommendations	10.1 Align with productivity goals 10.2 Cost efficiency 10.3 Quality standards 10.4 Safety requirements

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify major production stages and processes in leather goods and footwear manufacturing. 1.2. Interpret current workflow and material movement to identify inefficiencies and bottlenecks. 1.3. Understand methods for monitoring delays, idle times, and workflow interruptions. 1.4. Identify techniques for recording process timings, material flow, and handling accurately. 1.5. Interpret collected production data to detect causes of delays, waste, and inefficiency. 1.6. Identify areas and strategies for process improvement to reduce waste and enhance efficiency. 1.7. Interpret how improvement recommendations align with overall productivity and organizational goals.
2. Underpinning Skills	2.1. Performing identification of key production stages in leather and footwear manufacturing. 2.2. Applying observation skills to monitor workflow, material movement, and process efficiency. 2.3. Applying analytical thinking to identify bottlenecks, delays, and waste in production. 2.4. Performing root cause analysis to determine factors affecting productivity. 2.5. Applying problem-solving skills to develop practical improvement suggestions. 2.6. Demonstrating documentation and reporting skills for accurate data and findings. 2.7. Applying communication skills to share improvement ideas with relevant personnel.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Internet 4.4 Projector 4.5 Stationary 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified key production stages in leather and footwear manufacturing. 1.2 assessed current workflow and identify bottlenecks. 1.3 recorded process timings and material flow. 1.4 analysed causes of production delays and waste. 1.5 interpreted areas for improvement.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY BASIC INDUSTRIAL ENGINEERING (IE) ACTIVITIES	Nominal Duration: 70 Hrs.	Unit Code: SICIP-LET-IP1-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply industrial engineering (IE) activities. It specifically includes the tasks of identifying different types of line layout, performing work study process, preparing skills matrix for production line, implementing line balancing in production, developing value stream mapping and preparing efficiency report.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify different types of line layout.	1.1 <u>Different types of production line layouts</u> are identified and listed. 1.2 Features of each layout type are observed and recorded. 1.3 Advantages and disadvantages of various line layouts are analyzed. 1.4 Layout suitability is evaluated based on product type and workflow. 1.5 Existing layouts are compared with recommended standards. 1.6 Selected line layout is implemented in the production area.
2. Perform work study process.	2.1 <u>Work study process</u> is performed to analyze and improve productivity. 2.2 Method study is conducted to identify the most efficient way of performing tasks. 2.3 Current work methods are observed, recorded, and evaluated. 2.4 <u>Time study</u> is carried out to measure task duration accurately. 2.5 Standard time for each operation is calculated and documented. 2.6 Findings from method and time studies are analyzed for process improvement. 2.7 Results are reported and used to establish performance benchmarks.
3. Prepare skills matrix for production line	3.1 <u>Skills variables</u> required for the production line are identified and listed. 3.2 Skill weights are determined based on task importance and complexity. 3.3 Skill levels and grades are defined according to standards.

	3.4 Multi-skilled criteria are identified. 3.5 Data on worker competencies are collected and recorded. 3.6 Skills matrix is prepared showing workers' proficiency across tasks. 3.7 Completed skills matrix is reviewed and updated for accuracy and use in workforce planning.
4. Implement line balancing in production.	4.1 Principles and objectives of line balancing are understood and explained. 4.2 Work tasks and process steps are analyzed for balance and efficiency. 4.3 <u>Standard line balancing methods</u> are applied to distribute workload evenly. 4.4 Real-time line balancing adjustments are implemented during production. 4.5 Idle time and bottlenecks are identified and minimized. 4.6 Production flow is monitored to ensure smooth operation. 4.7 Line balancing results are reviewed and documented for continuous improvement.
5. Develop value stream mapping.	5.1 The concept and purpose of Value Stream Mapping (VSM) are understood and explained. 5.2 Current production processes are analyzed to identify value-adding and non-value-adding activities. 5.3 <u>Appropriate VSM level</u> is determined based on process scope and objectives. 5.4 Data on material flow, information flow, and process time are collected. 5.5 Value Stream Map is prepared to visualize the entire workflow. 5.6 Areas for improvement are identified through the completed VSM. 5.7 The finalized VSM is reviewed and used for process optimization planning.
6. Prepare efficiency report	6.1 Data on machine performance and production output are collected accurately. 6.2 Overall Equipment Effectiveness (OEE) is calculated based on availability, performance, and quality. 6.3 Overall Working Efficiency (OWE) is measured to evaluate labor and process productivity. 6.4 Collected data are analyzed to identify performance gaps and improvement areas. 6.5 Efficiency results are summarized and documented in a standard report format. 6.6 <u>Reports</u> are reviewed with relevant personnel for validation and decision-making. 6.7 Recommendations for improving OEE and OWE are

	included in the final efficiency report.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Different types of production line layouts	1.1 Product layout 1.2 Process layout 1.3 Hybrid layout 1.4 Fixed position layout
2. Work study process	2.1 Time study 2.2 Method study
3. Time study	3.1 Manual 3.2 Stop watch 3.3 Pre-determined Motion Time System (PMTS)
4. Skills variables	4.1 Standard time 4.2 Quality
5. Standard line balancing methods	5.1 Standard 5.2 Real time
6. Appropriate VSM level	6.1 Current VSM 6.2 Future VSM
7. Reports	7.1 Manual (Spreadsheet) 7.2 Software

Curricular Content Guide

1. Underpinning Knowledge	1.1. Identify different types of production line layouts and understand their features, advantages, and limitations. 1.2. Interpret layout suitability based on product type, workflow, and production requirements. 1.3. Understand the purpose and process of work study, including method study and time study. 1.4. Identify key skill variables and develop a skills matrix. 1.5. Interpret techniques to identify and minimize idle time and bottlenecks for smooth production flow. 1.6. Understand the concept and application of Value Stream Mapping (VSM). 1.7. Identify methods for collecting and analyzing production and performance data.
2. Underpinning Skills	2.1. Demonstrating understanding of different types of production line layouts and their applications. 2.2. Applying analytical skills to evaluate layout suitability, efficiency, and workflow balance. 2.3. Performing work study techniques. 2.4. Applying mathematical and observation skills to calculate standard time and analyze processes. 2.5. Demonstrating data management skills to develop and maintain an accurate skills matrix.

	2.6. Applying line balancing principles. 2.7. Performing analysis of value-adding and non-value-adding activities using Value Stream Mapping (VSM). 2.8. Applying problem-solving skills.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Internet 4.4 Projector 4.5 Stationary 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified different types of line layout. 1.2 performed work study process. 1.3 prepared skills matrix for production line 1.4 implemented line balancing in production. 1.5 developed value stream mapping. 1.6 prepared efficiency report
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY LEAN MANUFACTURING TECHNIQUES	Nominal Duration: 40 Hrs.	Unit Code: SICIP-LET-IPI-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply lean manufacturing techniques. It specifically includes the tasks of identifying 3M (muda, muri and mura), identifying seven types of waste in production process, applying 5S techniques in workplace, implementing lean tools and implementing continuous improvement (KAIZEN) activities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify 3M (muda, muri and mura)	1.1 Activities causing Muda (waste) in the production process are identified and documented. 1.2 Tasks leading to Muri (overburden) on workers or machines are detected and recorded. 1.3 Variations and inconsistencies causing Mura (unevenness) in workflow are recognized. 1.4 Observations are compared with standard operating practices.
2. Identify the seven types of waste in production process.	2.1 <u>Seven types of waste in the production process</u> are identified and listed. 2.2 Each type of waste is observed in the workflow and documented. 2.3 <u>Causes of waste</u> in materials, time, and labor are recognized. 2.4 Instances of defects and rework are detected and recorded. 2.5 Waste related to transportation, motion, and inventory is analyzed.
3. Apply 5S techniques in workplace.	3.1 Workplace is organized according to the <u>5S principles</u> . 3.2 Unnecessary items are removed and sorted from the work area. 3.3 Tools and materials are arranged in designated locations for easy access. 3.4 Workstations are cleaned and maintained regularly. 3.5 Standard procedures for organization and cleanliness are implemented. 3.6 5S practices are monitored and deviations are corrected.
4. Implement lean tools	4.1 <u>Lean tools</u> are implemented in the production process according to requirements.

	4.2 Just-in-Time techniques are applied and monitored. 4.3 Jidoka principles are applied in the workflow. 4.4 Production activities are aligned with lean principles to reduce waste and improve efficiency. 4.5 Deviations from lean practices are identified and corrective measures are documented. 4.6 Implementation results are reviewed against productivity and quality targets.
5. Implement continuous improvement (Kaizen) activities.	5.1 <u>Kaizen activities</u> are implemented in the workplace to improve processes. 5.2 Improvement opportunities are identified through observation and feedback. 5.3 Small, incremental changes are applied to enhance efficiency and quality. 5.4 Results of improvement actions are measured and documented. 5.5 Employees are involved in suggesting and implementing improvements. 5.6 Outcomes of Kaizen activities are reviewed against performance targets.

Range of Variables

Variable	Range (Includes but not limited to):
1. Seven types of waste in the production process	1.1 Transportation 1.2 Inventory 1.3 Motion 1.4 Waiting 1.5 Overproduction 1.6 Overprocessing 1.7 Defects
2. Causes of waste	2.1. Poor planning 2.2. Machine downtime 2.3. Operator errors 2.4. Skill gaps 2.5. Layout inefficiencies 2.6. Material shortage
3. 5S principles	3.1. Sort 3.2. Set in order 3.3. Shine 3.4. Standardize 3.5. Sustain
4. Lean tools	4.1 Just-in-time 4.1.1 Takt time 4.1.2 One pieces flow 4.1.3 Line balancing

	4.1.4 Single Minute Exchange of Dies (SMED) 4.1.5 KANBAN 4.2 Jidoka 4.2.1 ANDON system 4.2.2 POKA YOKE 4.2.3 Skills matrix 4.2.4 Autonomation 4.2.5 Total Productivity Maintenance (TPM) 4.2.6 Visual management
5. Kaizen activities	5.1 Process simplification 5.2 Workstation reorganization 5.3 Standardization of tasks 5.4 Minor equipment adjustments 5.5 Problem identification and solution

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the concepts of 3M-Muda (waste), Muri (overburden), and Mura (unevenness). 1.2 Interpret the seven types of waste in production. 1.3 Understand methods to detect and record waste related. 1.4 Identify and apply 5S workplace organization principles. 1.5 Interpret the importance of cleanliness, order, and standardization. 1.6 Understand lean manufacturing tools and their application. 1.7 Identify deviations from lean practices and determine corrective actions for process improvement. 1.8 Interpret the concept of continuous improvement (Kaizen).
2. Underpinning Skills	2.1. Demonstrating understanding of the 3M concepts (Muda, Muri, Mura) and their impact on productivity. 2.2. Applying observation and analytical skills. 2.3. Performing evaluation of workflow. 2.4. Applying 5S techniques to organize, clean, and maintain an efficient and safe workplace. 2.5. Demonstrating problem-solving skills. 2.6. Applying Just-in-Time and Jidoka principles. 2.7. Performing continuous monitoring. 2.8. Demonstrating teamwork and communication skills. 2.9. Applying evaluation and reporting skills.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified 3M (muda, muri and mura) 1.2 identified seven types of waste in production process. 1.3 applied 5S techniques in workplace. 1.4 implemented lean tools 1.5 implemented continuous improvement (Kaizen) activities.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPTIMIZE MACHINE AND EQUIPMENT UTILIZATION	Nominal Duration: 40 Hrs.	Unit Code: SICIP-LET-IP1-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to optimize machine and equipment utilization. It specifically includes tasks of identifying machines in the production line, recording machine downtime and its causes, scheduling regular inspection and implementing quick changeover (SMED) techniques.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify machines in the production line.	1.1 <u>Machines used in the leather goods production line</u> are identified and listed. 1.2 <u>Machines used in footwear production line</u> are identified and listed. 1.3 Each machine is observed for its function and operation. 1.4 Machine types and specifications are recorded accurately. 1.5 Location of machines within the production line is documented. 1.6 Safety features and operating procedures of machines are noted.
2. Record machine downtime and its causes.	2.1 Machine downtime is recorded systematically during production. 2.2 Start and end times of each downtime event are documented. 2.3 <u>Causes of downtime</u> are identified. 2.4 Frequency and duration of machine stoppages are analyzed. 2.5 Observations are compared with maintenance schedules and standards. 2.6 Downtime data are documented using approved formats or tools. 2.7 Reports are prepared to support corrective and preventive actions.
3. Schedule regular inspection.	3.1 Regular inspection schedules for machines are prepared and documented. 3.2 Inspection intervals are determined based on manufacturer guidelines and production needs. 3.3 <u>Inspection activities (CLITA)</u> are planned and communicated to relevant personnel. 3.4 Machines are checked according to the scheduled plan.

	3.5 Inspection results, including any defects or issues, are recorded. 3.6 Deviations from normal operation are reported for corrective action. 3.7 Inspection schedules are reviewed and updated as needed to ensure reliability.
4. Implement quick changeover (SMED) techniques.	4.1 <u>SMED (Single Minute Exchange of Die) techniques</u> are implemented on relevant machines. 4.2 Current changeover processes are observed and recorded. 4.3 <u>Internal and external setup activities</u> are identified and separated. 4.4 Time required for each changeover step is measured and documented. 4.5 <u>SMED steps</u> are reorganized to minimize downtime and improve efficiency. 4.6 Quick changeover methods are applied and monitored for effectiveness.

Range of Variables

Variable	Range (Includes but not limited to):
1. Machines used in the leather goods production line	1.1 Cutting machine 1.2 Skiving machine 1.3 Stitching/sewing machine 1.4 Edge creasing machine 1.5 Edge coloring machine 1.6 Embossing machine 1.7 Riveting/press machine 1.8 Gluing/adhesive application machine 1.9 Polishing/buffing machine 1.10 Packaging machine
2. Machines used in footwear production line	2.1 Cutting machine 2.2 Skiving machine 2.3 Stitching/sewing machine 2.4 Lasting machine 2.5 Sole pressing/attaching machine 2.6 Edge trimming machine 2.7 Finishing/polishing machine 2.8 Embossing or branding machine 2.9 Riveting/press machine 2.10 Packaging machine
3. Causes of downtime	3.1 Mechanical faults 3.2 Electrical failures 3.3 Operator errors 3.4 Tool wear 3.5 Lack of materials
4. Inspection activities	4.1 Visual checks

(CLITA)	4.2 Operational tests 4.3 Lubrication 4.4 Alignment 4.5 Safety checks
5. SMED (Single Minute Exchange of Die) techniques	5.1 Separating internal and external setup 5.2 Parallel operations 5.3 Standardizing tools 5.4 Pre-preparation of materials
6. Internal and external setup activities	6.1 Internal: tasks done while the machine is stopped. 6.2 External: tasks done while the machine is running.
7. SMED steps	7.1 Separate internal and external setup 7.2 Pre-prepare tools and materials 7.3 Standardize procedures 7.4 Convert internal activities to external activities 7.5 Sequence tasks efficiently 7.6 Automation of activities 7.7 Parallelize operations where possible 7.8 Eliminate unnecessary movements 7.9 Test and adjust quickly

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify machines used in leather goods and footwear production lines. 1.2 Interpret the layout, location, and operating procedures of machines within the production process. 1.3 Understand methods for recording machine downtime, including start and end times, causes, and frequency. 1.4 Analyze downtime data to identify trends and support corrective and preventive maintenance actions. 1.5 Identify procedures for scheduling and conducting regular machine inspections. 1.6 Interpret the CLITA (Clean, Lubricate, Inspect, Tighten, Adjust) approach. 1.7 Understand the purpose and application of SMED (Single Minute Exchange of Die) for quick changeover.
2. Underpinning Skills	2.1 Applying observation skills to understand machine functions, specifications, and operating procedures. 2.2 Demonstrating recordkeeping and data management skills to track machine locations and performance. 2.3 Performing accurate recording and analysis of machine downtime and its causes. 2.4 Applying analytical and problem-solving skills to identify trends and propose maintenance solutions. 2.5 Demonstrating technical understanding of CLITA and SMED techniques for machine efficiency. 2.6 Applying time management and process analysis skills.

	2.7 Demonstrating teamwork and communication skills.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified machines in the production line. 1.2 recorded machine downtime and its causes. 1.3 scheduled regular inspection. 1.4 implemented quick changeover (SMED) techniques.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPROVE WORKFORCE EFFICIENCY	Nominal Duration: 30 Hrs.	Unit Code: SICIP-LET-IPI-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to improve workforce efficiency. It specifically includes the tasks of identifying skill levels of workers, setting realistic production targets, assigning tasks based on worker skills capacity and monitoring performance and provide feedback.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify skill levels of workers.	1.1 Workers are categorized according to proficiency levels. 1.2 Workers' skill levels are identified using a <u>skills matrix</u> . 1.3 Competencies required for each role are documented and compared with worker abilities. 1.4 Observations and assessments are recorded systematically. 1.5 Gaps in skills and knowledge are detected and noted. 1.6 Skills matrix is updated regularly to reflect current capabilities.
2. Set realistic production targets.	2.1 <u>Production targets</u> are set based on analysis of current capacity and resources. 2.2 <u>Historical production data and performance trends</u> are reviewed. 2.3 Targets are aligned with quality and efficiency standards. 2.4 Realistic timelines for achieving production goals are established. 2.5 Workforce capabilities and machine availability are considered. 2.6 Targets are periodically reviewed and adjusted based on performance feedback.
3. Assign tasks based on worker skills capability.	3.1 Tasks are assigned according to workers' skill levels and competencies. 3.2 Worker capabilities are reviewed using the skills matrix. 3.3 Job responsibilities are matched with proficiency and experience. 3.4 Assignments are documented and communicated clearly. 3.5 Task allocation is monitored to ensure balanced workload.

	3.6 Adjustments are made based on performance and observed efficiency.
4. Monitor performance and provide feedback.	4.1 Worker performance is monitored regularly against set targets. 4.2 <u>Observations of task execution</u> are recorded and analyzed. 4.3 Deviations from expected performance standards are identified. 4.4 Feedback is provided to workers in a constructive manner. 4.5 <u>Areas for improvement</u> are highlighted and documented. 4.6 Progress after feedback is tracked and reviewed.

Range of Variables

Variable	Range (Includes but not limited to):
1. Skills Matrix	1.1 Technical skills 1.2 Operational abilities 1.3 Safety knowledge 1.4 Quality standards
2. Production targets	2.1 Hourly 2.2 Daily 2.3 Weekly 2.4 Monthly
3. Historical production data and performance trends	3.1 Previous production records 3.2 Defect rates 3.3 Machine utilization 3.4 Operator performance
4. Observations of task execution	4.1 Conducted through direct supervision 4.2 Checklists 4.3 Time studies 4.4 Video monitoring
5. Areas for improvement	5.1 Skill gaps 5.2 Process understanding 5.3 Speed 5.4 Accuracy 5.5 Teamwork

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify methods for categorizing workers. 1.2 Understand how to use and update a skills matrix. 1.3 Interpret procedures for identifying skill gaps and recording assessment results. 1.4 Identify factors influencing production goals. 1.5 Interpret methods for assigning tasks based on worker skills, experience, and proficiency levels.
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	1.6 Interpret techniques for providing constructive feedback and supporting continuous improvement. 1.7 Identify ways to track worker progress and evaluate performance improvement over time.
2. Underpinning Skills	2.1. Performing assessment and categorization of workers based on skill levels and competencies. 2.2. Applying observation and evaluation skills to identify gaps between required and existing abilities. 2.3. Applying analytical skills to set realistic and achievable production targets. 2.4. Demonstrating communication skills to provide clear instructions and constructive feedback. 2.5. Applying monitoring and performance analysis skills to track progress and identify improvement areas. 2.6. Applying coaching and mentoring skills to support continuous workforce development and efficiency.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified skill levels of workers. 1.2 set realistic production targets. 1.3 assigned tasks based on worker skills capability. 1.4 monitored performance and provide feedback.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified

	assessor or occupation-specific industry expert.
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Unit of Competency: IMPLEMENT QUALITY CONTROL AND ASSURANCE TECHNIQUES	Nominal Duration: 20 Hrs.	Unit Code: SICIP-LET-IPI-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to implement quality control and assurance techniques. It specifically includes the tasks of identifying common defects, inspecting materials and products, applying standard operating procedures (SOPs) and recording and reporting quality deviations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify common defects.	1.1 <u>Common defects in leather goods</u> are identified and documented. 1.2 <u>Common defects in footwear products</u> are identified and documented. 1.3 Defective items are inspected visually and compared with quality standards. 1.4 Causes of defects related to materials, machines, or workmanship are recorded. 1.5 Defects are classified according to severity and production stage. 1.6 Findings are reported to relevant personnel for corrective action.
2. Inspect materials and products.	2.1 Materials used in production are identified and inspected for quality. 2.2 Products at various stages are inspected against specifications and standards. 2.3 Defective or non-conforming materials and products are detected and recorded. 2.4 Inspection results are documented accurately for traceability. 2.5 Observations are compared with quality benchmarks and standards. 2.6 Issues detected during inspection are reported to relevant personnel.
3. Apply standard operating procedures (SOPs).	3.1 Standard Operating Procedures (SOPs) are understood and explained. 3.2 SOPs are followed consistently during all relevant work activities. 3.3 Work processes are performed in compliance with established SOPs. 3.4 Deviations from SOPs are identified and reported. 3.5 SOPs are maintained and updated as required. 3.6 Compliance with SOPs is monitored and documented.

4. Record and report quality deviations.	4.1 Basic quality tools are identified and used for inspection purposes. 4.2 Quality deviations in materials and products are recorded accurately. 4.3 Records are maintained according to established procedures. 4.4 Deviations from quality standards are documented with relevant details. 4.5 Quality reports are generated systematically and reviewed. 4.6 Reports are submitted to relevant personnel for corrective action.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Common defects in leather goods	1.1 Cutting Department 1.1.1 Misaligned patterns 1.1.2 Uneven or inaccurate cuts 1.1.3 Wastage of leather 1.1.4 Scratches or cuts on leather surface 1.2 Skiving/Preparation Department 1.2.1 Uneven skiving thickness 1.2.2 Torn or weakened edges 1.2.3 Wrinkles or creases on leather 1.3 Stitching/Sewing Department 1.3.1 Loose, skipped, or uneven stitches 1.3.2 Thread breakage 1.3.3 Misalignment of parts during sewing 1.4 Assembling/Adhesive Department 1.4.1 Glue marks on visible surfaces 1.4.2 Improper bonding of components 1.4.3 Warping or deformation after adhesive application 1.5 Finishing Department 1.5.1 Uneven coloring or staining 1.5.2 Surface cracks or scratches 1.5.3 Poor edge finishing 1.5.4 Improper embossing or stamping
2 Common defects in footwear products	2.1 Cutting Department 2.1.1 Misaligned patterns or templates 2.1.2 Uneven or inaccurate cuts 2.1.3 Leather or fabric scratches 2.1.4 Wastage of material 2.2 Stitching/Sewing Department 2.2.1 Loose, skipped, or uneven stitches 2.2.2 Thread breakage 2.2.3 Misalignment of upper components

	2.3 Lasting Department 2.3.1 Poor shaping of the upper over the last 2.3.2 Uneven tension or wrinkles 2.3.3 Misalignment between upper and sole 2.4 Sole Preparation and Attaching Department 2.4.1 Uneven sole thickness 2.4.2 Improper bonding or sole detachment 2.4.3 Glue marks on visible surfaces 2.5 Finishing Department 2.5.1 Surface scratches, scuffs, or cracks 2.5.2 Color inconsistency or staining 2.5.3 Poor edge finishing or polishing
3 Basic quality tools	3.1 Check Sheets 3.2 Control Charts 3.3 Pareto Analysis 3.4 Cause-and-Effect (Fishbone/Ishikawa) Diagram 3.5 Flowcharts 3.6 Histogram 3.7 Scatter Diagram 3.8 5 Whys Analysis 3.9 Process Mapping 3.10 Inspection Gauges and Measuring Tools (rulers, calipers, micrometers)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify common defects in leather goods and footwear. 1.2 Understand the causes of defects related to materials, machinery, and workmanship. 1.3 Identify quality inspection techniques for materials and products at different production stages. 1.4 Understand the purpose and application of Standard Operating Procedures (SOPs) in maintaining quality. 1.5 Identify deviations from SOPs and report them for corrective action and process improvement. 1.6 Understand the use of basic quality tools for inspection and data recording. 1.7 Interpret procedures for documenting, reporting, and reviewing quality deviations.
2. Underpinning Skills	2.1 Performing visual inspection to identify and classify common defects in leather goods and footwear. 2.2 Applying analytical skills to determine defect causes related to materials, machines, or workmanship. 2.3 Applying knowledge of Standard Operating Procedures (SOPs). 2.4 Performing documentation and recordkeeping of inspection results and quality deviations. 2.5 Applying problem-solving skills to report and support

	corrective actions for identified defects. 2.6 Demonstrating use of basic quality tools for inspection, analysis, and reporting. 2.7 Performing continuous monitoring.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified common defects. 1.2 inspected materials and products. 1.3 applied standard operating procedures (SOPs). 1.4 recorded and reported quality deviations.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MONITOR AND EVALUATE PRODUCTIVITY PERFORMANCE	Nominal Duration: 30 Hrs.	Unit Code: SICIP-LET-IPI-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to monitor and evaluate productivity performance. It specifically includes the tasks of collecting data on production, calculating key productivity indicators, comparing performance against set targets, identifying causes of underperformance and preparing reports and suggest improvement actions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collect data on production.	1.1 <u>Production data requirements</u> are identified and documented. 1.2 Data on output, labor hours, and machine utilization are collected accurately. 1.3 Standardized <u>data collection methods and tools</u> are applied. 1.4 Collected data is verified for accuracy and completeness. 1.5 <u>Variations between planned and actual production</u> are recorded. 1.6 Data is organized systematically for analysis and reporting.
2. Calculate key productivity indicators.	2.1 <u>Key productivity indicators</u> relevant to the leather and footwear sector are identified. 2.2 <u>Data on input and output variables</u> is analyzed for calculation. 2.3 Labor, machine, and material productivity are computed using <u>standard formulas</u> . 2.4 Calculated indicators are compared with industry benchmarks or targets. 2.5 Deviations in productivity performance are identified and recorded. 2.6 Calculations are verified to ensure accuracy and reliability.
3. Compare performance against set targets.	3.1 <u>Set productivity targets and benchmarks</u> are reviewed and documented. 3.2 Actual performance data is compared with planned targets. 3.3 Variations and gaps in performance are identified and analyzed. 3.4 <u>Causes of underperformance or improvement</u> are

	recorded. 3.5 Comparison results are summarized and reported to relevant personnel. 3.6 Corrective and preventive measures are suggested based on findings.
4. Identify causes of underperformance.	4.1 Areas of underperformance are identified through analysis of productivity data. 4.2 <u>Contributing factors</u> are examined. 4.3 Data from production and quality records is reviewed to detect inefficiencies. 4.4 Root causes of low performance are analyzed using <u>appropriate tools and techniques</u>. 4.5 Causes of underperformance are documented systematically.
5. Prepare reports and suggest improvement actions.	5.1 Reports are reviewed and approved by management or relevant personnel. 5.2 Recommendations are communicated to concerned departments for implementation. 5.3 Records of reports and improvement actions are maintained for follow-up and evaluation.

Range of Variables

Variable	Range (Includes but not limited to):
1. Production data requirements	1.1 Data types 1.1.1 Output 1.1.2 Downtime 1.1.3 Defects 1.1.4 Material usage 1.2 Data sources 1.2.1 Operators 1.2.2 Machines 1.2.3 Supervisors 1.3 Data collection frequency 1.3.1 Hourly 1.3.2 Daily 1.3.3 Weekly
2. Data collection methods and tools	2.1 Collection methods 2.1.1 Manual 2.1.2 Digital 2.1.3 Automated 2.2 Data collection tools 2.2.1 Checklists 2.2.2 Logbooks 2.2.3 Apps 2.2.4 Software
3. Variations between	3.1 Planned vs. actual output

planned and actual production	3.2 Planned vs. actual labor hours 3.3 Planned vs. actual machine uptime 3.4 Causes of variation 1.4.1 Machine breakdown 1.4.2 Material delay 1.4.3 Absenteeism 3.5 Deviation percentage 3.6 Reporting schedule
4. Key productivity indicators	4.1 Labor productivity 4.2 Machine productivity 4.3 Material productivity 4.4 Overall equipment effectiveness (OEE) 4.5 Output per hour or shift 4.6 Defect or rejection rate
5. Data on input and output variables	5.1 Input variables 5.1.1 Labor hours 5.1.2 Machine hours 5.1.3 Material usage 5.2 Output variables 5.2.1 Finished pairs 5.2.2 Units produced 5.2.3 Usable output 5.3 Production time 5.3.1 Cycle time 5.3.2 Downtime 5.3.3 Setup time 5.4 Work shifts and batch sizes
6. Standard formulas	6.1 Labor productivity = Output / Labor hours 6.2 Machine productivity = Output / Machine hours 6.3 Material productivity = Output / Material used 6.4 OEE = Availability × Performance × Quality
7. Set productivity targets and benchmarks	7.1 Planned output targets 7.1.1 Daily 7.1.2 Weekly 7.1.3 Monthly 7.2 Standard productivity levels 7.2.1 Labor 7.2.2 Machine 7.2.3 Material 7.3 Industry or enterprise benchmarks 7.4 Historical performance data 7.5 Customer or buyer standards
8. Causes of underperformance or improvement	8.1 Machine downtime 8.2 Low worker efficiency 8.3 Poor material quality 8.4 Unbalanced workflow or delays 8.5 Improved process or training

9. Contributing factors	9.1 Manpower issues 9.1.1 Absenteeism 9.1.2 Skill gaps 9.1.3 Fatigue 9.2 Material issues 9.3 Machinery issues
10. Appropriate tools and techniques	10.1 Fishbone (Ishikawa) diagram 10.2 5 Whys analysis 10.3 Pareto analysis 10.4 Flow charting or process mapping 10.5 Cause and effect matrix

Curricular Content Guide

1. Underpinning Knowledge	1.1 identify production data requirements 1.2 understand methods for accurate data collection and verification. 1.3 Interpret standardized tools and techniques for gathering and organizing production data. 1.4 Identify methods to compare actual performance against planned targets and benchmarks. 1.5 Interpret deviations in productivity and analyze causes of underperformance using relevant data. 1.6 Understand the process of reviewing production and quality records to detect inefficiencies. 1.7 Identify root causes of low performance using analytical tools and techniques. 1.8 Interpret how to prepare productivity and performance reports for management review.
2. Underpinning Skills	2.1 Performing accurate collection and organization of production data using standard methods and tools. 2.2 Applying analytical and mathematical skills to calculate key productivity indicators. 2.3 Demonstrating attention to detail in verifying the accuracy and completeness of collected data. 2.4 Applying critical thinking to compare actual performance against set targets and benchmarks. 2.5 Performing root cause analysis to identify factors contributing to underperformance. 2.6 Demonstrating proficiency in preparing clear, data-driven productivity and performance reports. 2.7 Performing recordkeeping and documentation to ensure traceability and support continuous improvement.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected data on production. 1.2 calculated key productivity indicators. 1.3 compared performance against set targets. 1.4 identified causes of underperformance. 1.5 prepared reports and suggest improvement actions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Industrial Productivity Improvement
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	01
2.	Multimedia projector & screen	01
3.	Scanner	01
4.	Printer	01
5.	Stopwatch	01
6.	Time study timer	05
7.	Measuring tape	20
8.	Line layout model	05
9.	Work study table	02
10.	Calculator	10
11.	Kaizen Boards	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Industrial Productivity Improvement the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.